

a few of the problems your system faces every day. Let's find out how PCI Express will save us.

On The PCI Expressway

The most interesting thing to note is the legacy PCI bus hanging at the bottom right of the infographic (below). PCI becomes a subset of PCIe. To understand how PCIe betters this, let's first understand the basic structure of PCIe.

Unlike PCI, PCIe is a two-lane transfer system. This lets data come and go simultaneously. In PCIe, each of the dedicated connections/paths between the switch and the device is called a link. A PCIe x1 link has one lane, which can transfer data to and fro simultaneously, providing transfer speeds of 2.5 GB/s one way, such that an x1 link has a total transfer capacity of 5 GB/s.

Now, if we add another lane to this, we get an x2 link, wherein each lane can transfer 5 GB/s, making the total available bandwidth 10 GB/s. We can theoretically have x32 PCIe connections with transfer speeds of 20 GB/s.

To put things into perspective, if you could manage a x32 PCIe connection between, say, your 80 GB HDD to another 80 GB HDD, you could transfer the entire 80 GB of data in about four seconds! But how does all this bandwidth help us?

Remember how in PCI, if the processor was talking to the Northbridge, it was inaccessible to other elements of a system due to the single-channel nature of PCI? The dual-channel feature of PCIe overcomes this and allows multiple devices to communicate with each other.

PCIe will replace the entire bus architecture on a motherboard and would be *the* interconnect between various devices and bridges as well as *the* interconnect between the bridges.

PCIe is incredibly scalable and allows additional links to be provided as and when bandwidth requirements increase. For example, for a Gigabit LAN card, an x1 lane is sufficient, but for replacing your 8X AGP card, a motherboard burning an x16 link is necessary and possible.

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Another aspect that has changed in PCIe is the legacy concept of PCI bus mastering. PCIe features a switch system that sits in the centre, and all the system devices connect to it via a dedicated link similar to the bus connecting the CPU to the Northbridge. This is called a 'Point-to-Point Interconnect System'.

What this switch does is route all the PCI/PCIe data through itself, allowing two devices to talk to each other—if the communication does not involve any cache-specific data or instructions, this switch will simply offer the two devices a dedicated and direct link to share with each other. This obviously makes information exchange between devices extremely fast.

What Does This Mean?

The question now is, how well does all this translate to real-world usage. Will your games finally run enormously faster and better?

Well, not really. The performance difference between AGP 4x (1 GB/s) and AGP 8x (2 GB/s) is negligible. So, PCIe x16 cards will not make Doom 3 run any faster because there simply isn't that much data to transfer.

Immediate beneficiaries would be people into video-editing; the performance of their machines would see a huge boost due to the dual-channel capability. This is because peripheral devices such as hard disks or external FireWire devices will be able to reap the benefits that additional bandwidth and reduced latencies offer.

Our recommendation is that you should not upgrade to PCIe unless you work heavily on video editing. If, however, you are about to upgrade anyway, go ahead and get a PCIe-based motherboard—PCIe as a standard is an inevitability.

A word of caution, though: a motherboard can only support either AGP or PCIe. So if you buy a PCIe motherboard, you will have to buy a PCIe graphics card as well. On the upside, PCIe is backward-compatible with older PCI components.

The PCI Versus PCIe Architecture

